

Cryptogrphy

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Vedio1	https://youtube.com/playlist?list=PL60oESahDerr25n97ugHfJKzQ7SjSjVH&si=MGQD3WUsVSVoeYGu

Basics :

Hasing:

RSA:

- Sub case

MD5:

Unknown Hash Detect :

John the ripper

John the ripper (two)

Multi Level Encryption

lets talk about Crypto basic :

Basic Term Of Cryptography

- Symmetric Cryptography:
 - a. **Caesar Cipher**
 - b. Rot13
 - c. **Vigenere Cipher**
 - d. Morse code
 - e. Bacon Cipher
 - f. Alphabetical substitution

###Tools:

- a. Cyberchef
- b. Caesar Cipher
- c. Vigenere Solver

Basic Term Of Cryptography

► Hash Function Cryptography:

- a. **Md5**
- b. SHA-1
- c. **SHA-256**
- d. Whirpool
- e. Shake
- f. HAS-160

###Tools:

- a. Cyberchef
- b. Hashcat (Default on Kali Linux OS)
- c. John the ripper (Default On Kali Linux OS)
- d. Online Tools(Search :Online hash decoder tools)

Basic term Of Cryptography

► Asymmetric Cypto:

- i. Use public and private key
- ii. public key use for encryption
- iii. private key use for decryption

Example: RSA Algorithm

Tools :

- a. openssl (Default on Kali Linux OS)
- b. Online tools
- c. Python code

Basic Term Of Cryptography

► Encoder System:

- a. **Base64**
- b. binary
- c. **Base32**
- d. Hexadecimal
- e. AES (Advance Encryption Standard)
- f. DES (Data Encryption Standard)

###Tools:

- a. Cyberchef
- b. Hashcat (Default on Kali Linux OS)
- c. Openssl (Default On Kali Linux OS)
- d. Online Tools(Search :Online hash decoder tools)

Tools :

<https://cryptii.com/pipes/caesar-cipher>

<https://medium.com/@anandrishav2228/cryptography-great-cheat-sheet-for-ctfs-d2ada754b319>

its here "/usr/share/wordlists/rockyou.txt"

Hasing :

```
-----Hashcat-----
Basic term :
-m -----hash type or hash mode like md5,sha-1 etc
-a -----attack mode or attack type like combination,dictionary,Brute-force
-o -----save output in file

Command:
hashcat -m hash mode -a attack mode -o save_outout target_file wordlist

Example :
if hash mode or hash type is md5 and select attack mode is Straight

hashcat -m 0 -a 0 -o result.txt md5hash_value /usr/shar/wordlist/rockyou.txt
```

```
Command:
hashcat -m hash mode -a attack mode -o save_outout target_file wordlist

Example :
if hash mode or hash type is md5 and select attack mode is Straight

hashcat -m 0 -a 0 -o result.txt md5hash_value /usr/shar/wordlist/rockyou.txt

----- Hash Value -----
1.48bb6e862e54f2a795ffc4e541caed4d9
2.CBFDAC6008F9CAB4083784CBD1874F76618D2A97
3.1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032

-----Hash with salt -----
hash: e5d8870e5bdd26602cab8dbe07a942c8669e56d6 salt :tryhackme (hash type -HMAC-sha1)
```

Basic term:

- m -----hash type or hash mode like md5,sha-1 etc
- a -----attack mode or attack type like combination,dictionary,Brute-force
- o -----save output in file

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Hash Value

1.48bb6e862e54f2a795ffc4e541caed4d

2.CBFDAC6008F9CAB4083784CBD1874F76618D2A97

3.1C8BFE8F801079745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032

Hash with salt

hash: e5d8870e5bdd26602cab8dbe07a942c8669e56d6

salt :*tryhackme* (hash type -HMAC-sha1)

--

`clear`

`ls`

`cat result.txt`

Check here for command :

Hash cats Cmds

সহজ ভাষায় বলি — **attack mode** নির্বাচন করার জন্য তোমাকে ৩টা প্রশ্ন নিজের কাছে করতে হবে। এই ৩টা প্রশ্নের উত্তর দেখলেই তুমি বুঝতে পারবে কোন **attack mode** ব্যবহার করতে হবে।

✓ ১) আমি কি জানি পাসওয়ার্ড **Wordlist**-এ থাকতে পারে?

যদি মনে হয় পাসওয়ার্ড

সাধারণ

dictionary-তে থাকতে পারে

rockyou বা অন্য তালিকায় থাকার সম্ভাবনা বেশি

➡ **Attack mode: -a 0 (Straight)**

এটাই সবসময় প্রথম চেষ্টা।

✓ ২) পাসওয়ার্ডের **pattern/structure** সম্পর্কে কিছু ধারণা আছে?

যেমন:

password 6 digit

শুধু সংখ্যা

শুধু small letters

শেষে 123 থাকে

format জানা (যেমন Name123)

➡ **Attack mode: -a 3 (Bruteforce / Mask)**

Example:

`hashcat -m 100 -a 3 hash.txt ?d?d?d?d?d`

✓ ৩) পাসওয়ার্ড **word + number / number + word**?

যদি মনে হয়

samin123

123samin

admin2024

hello99

ইত্যাদি

➡ **Hybrid attacks perfect:**

Ending numbers → -a 6

hashcat -m 100 -a 6 hash.txt rockyou.txt ?d?d?d

Starting numbers → -a 7

hashcat -m 100 -a 7 hash.txt ?d?d rockyou.txt

📌 মুখস্থ রাখার সহজ টেবিল

Password type

Attack Mode

Wordlist-e থাকতে পারে

- a 0

Format/pattern জানা

- a 3

Wordlist + number

- a 6

Number + wordlist

- a 7

দুই wordlist combine

- a 1

🔥 সবচেয়ে সহজ নিয়ম (Beginner friendly)

📌 সবসময় প্রথমে -a 0

📌 যদি না হয় → -a 0 --rules

📌 যদি password pattern বোঝা যায় → -a 3 mask

📌 যদি word + number বা number + word → -a 6 / -a 7

চাইলে তুমি আমাকে হ্যাশ, পাসওয়ার্ড সম্পর্কে ধারণা বা সিচুয়েশন বলো —

আমি সরাসরি বলে দিতে পারবো ঠিক কোন **attack mode** তোমার ক্ষেত্রে সবচেয়ে দ্রুত কাজ করবে।

MD5 :

`Nano word.txt` for open new word file

MD5 Hash Generator

Use this generator to create an MD5 hash of a string:

iloveislam

Generate →

Your String	iloveislam
MD5 Hash	5dfdfb89cb619faf9c7b1f52c09a9102 <button>Copy</button>
SHA1 Hash	783aa3e818e7ccc9304c3e83d010442ae17b54ef <button>Copy</button>

This MD5 hash generator is useful for encoding passwords, credit cards numbers and other sensitive data into MD5 hashes.

```
File Actions Edit View Help
(foysal@kali)-[~/Documents/python_code]
$ ls
ctf.py md5.py rsa-1.py RSA-Algorithm.py word.txt
(foysal@kali)-[~/Documents/python_code]
$ python3 md5.py
md5 hash: 5dfdfb89cb619faf9c7b1f52c09a9102
File name: word.txt
```

5dfdfb89cb619faf9c7b1f52c09a9102 final answer

Basic Command :

`man john` - all john functions

`locate john` : shows files of john

3 modes : single crack && wordlist && incremental ;

<https://www.tunnelsup.com/getting-started-cracking-password-hashes/>

Single :

User name upor base kore upper case lower case ...also alphabet add kore..first kaj ore

Wordlist :

Word list theke check

Increment :

Brute force attack line by line

Syntax :

```
john target_file
```

Example:

```
john hash.txt
```

Syntax :

```
john modes target_file
```

Example:

```
john --single hash.txt :
```

```
john --single f9d4049dd6a4dc35d40e5265954b2a46
```

So , first " nano john.txt " put the value check " cat john.txt " it shows value

Then run ...after check " john -show john.txt "showing if it cracked

Syntax :

```
john --wordlist hash.txt
```

```
john --incremental hash.txt
```

Example :

```
john --incremental john.txt
```

Best Command:

```
john --format=hash_type --wordlist=wordlist target_file
```

Best Command:

```
john --format=hash_type --wordlist=wordlist target_file
```

Example:

```
john --format = raw -md5 --wordlist =/usr/share/wordlist/rockyou.txt hash.txt
```

Works :

```
ha1ti b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3 it shows HC {Haschcat}then
```

```
nano hash1.txt then "b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3 " value save then
```

```
john --format=raw -sha1 --wordlist =/usr/share/wordlist/rockyou.txt hash.txt
```

```
john --format=raw-sha1 --wordlist=/usr/share/wordlists/rockyou.txt hash1.txt
```

Found : letmein answer flag

Hash:

1.f9d4049dd6a4dc35d40e5265954b2a46

2.b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3 -----

3.1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032

Note:

Cracked Hash store in pot path

/// john.pot ///

Part Two

First a Image name dekha.....na paile then

<https://www.duplichecker.com/reverse-image-search.php>

Via this Video, we will learn how to convert different cipher text into plain text.

Ruin Cipher :

Monk Cipher :

Bill Cipher :

Color Code Cipher :

- ----- Different Cipher -----

1. ruin cipher :

{odin_was_here}

2.monk cipher :

66 111 119 95 84 48 95 84 104 51 95 71 79 100

{Bow_T0_Th3_G0d}

3.color code :

{HELL_CS}

4. music note :

Challenge : RE O R LA MI M O N H LA DO K 3 RE

{D_O_R_A_E_M_O_N_H_A_C_K_3_D}

5.trouser cipher : Semaphore Cipher

{thankyouhacker}

6.bill cipher :

{mortylikessnakejazz}

Different Cipher - Google Drive

RSA :

```
from Crypto.Util.number import inverse
p = 61
q = 53
m = int(input("Enter value of m : "))
n = p * q
phi = (p-1)*(q-1)
e = 17
d = inverse(e, phi)
print("value of d : "+ str(d))
c = pow(m, e, n)
print("value of c : "+ str(c))
m = pow(c, d, n)
print("vlaue of m : "+ str(m))
```

- ----- RSA CTF Walkthrough -----

In this Video We explained How to solve RSA CTF Challenge Via Python .I Hope You will Like that .

Problem :

c=28767758880940662779934612526152562406674613203406706867456395986985664083182

n=73069886771625642807435783661014062604264768481735145873508846925735521695159

e=65537

- ----- Home Work Problem -----

N:

37415923547017213098893819652088052694795252162093236205030866324359578830858399212088135936525

e: 3

ciphertext (c):

2205316413931134031046440767620541984801091216351222789180582564557328762455422721368029531360076

Comment Your Flag):

SUB CASE :

Without prime number :

Base 64 rsa private key docker

Decoder web :

<https://www.devglan.com/online-tools/rsa-encryption-decryption>

Problem :

Problem Source : Shadow CTF 2021

nc :

NWv7jXhu8ygcxXwFhn31dk1Up8A1KN3xNvVkgcl4N3B4GZ6omJOkDHQpn4nhGIDPjhGE/YkA7aax+LrXQfkm1wWeZeZbA

pk :

MIICdglBADANBgkqhkiG9w0BAQEFAASCAmAwggJcAgEAAoGBAIL6su8Hg4Bdc3aG58ScJILPuurI6/HB4Mfz2HblyRSiXC

Last jodi "=" thake taile base 64

Unknown Hash Detect :

Tools Name And Video Content :

Hashid:

Haiti :

Hash analyzer

Hash-identification

1. `hashid -----linux`

example :

hashid hash_value

2. `hash-identifier ----- linux`

example :

hash-identifier

Hash_value

- *Haiti bestt.....

3. **haiti** ----- **linux** ----- **Favourite**

example :

haiti hashvalue

4. Hash-analyzer ----- online tools

How to install Haiti :

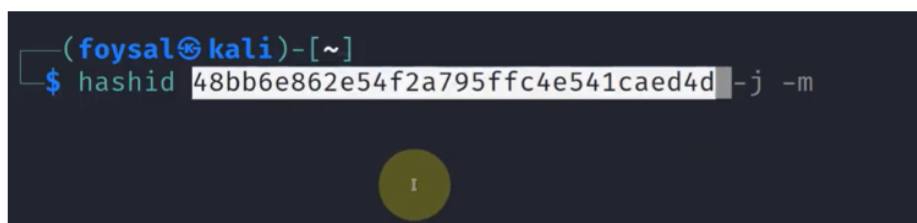
gem install haiti-hash

Hash:

1.48bb6e862e54f2a795ffc4e541caed4d ----- **easy**

2.CBFDAC6008F9CAB4083784CBD1874F76618D2A97 -----**password123**

3.1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032 ----**letmein**



```
(foysal@kali)-[~]  
$ hashid 48bb6e862e54f2a795ffc4e541caed4d -j -m
```

```
(foysal@kali)-[~]
$ hashid 48bb6e862e54f2a795ffc4e541caed4d
Analyzing '48bb6e862e54f2a795ffc4e541caed4d'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5
[+] Skype
```

man hashid

```
-m, --mode          include corresponding hashcat mode in output
-j, --john          include corresponding JohnTheRipper format in output
```

```
(foysal@kali)-[~]
$ hashid 48bb6e862e54f2a795ffc4e541caed4d -m
Analyzing '48bb6e862e54f2a795ffc4e541caed4d'
[+] MD2
[+] MD5 [Hashcat Mode: 0]
[+] MD4 [Hashcat Mode: 900]
[+] Double MD5 [Hashcat Mode: 2600]
[+] LM [Hashcat Mode: 3000]
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5 [Hashcat Mode: 8600]
[+] Skype [Hashcat Mode: 23]
```

Multi Level Encryption :

Via this Video we will learn how to crack multi level encryption cipher .

Using tools :

cyber chef

rapid table

Home work:

6b56e2c13c47d4e4529b7223473922c3

Hints:Bytersecsquad

MD5 > BrainFuck > Vegenere Decode with the Key > fuckinglife{ki_korba_jibone_bolo}

